

# Physics-Informed Neural Network Surrogate Modeling of Pressurized Cavity In Homogeneous and Bilayered Media

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**ABSTRACT:** The theory of cavity expansion can explain the mechanical response induced by the expansion or contraction of cylindrical cavities that underpin a wide range of geotechnical applications. While analytical methods often impose restrictive assumptions of homogeneity and isotropy, numerical approaches like the Finite Element Method (FEM) offer greater flexibility but can be computationally prohibitive. To address these limitations, we propose a novel Physics-Informed Neural Network (PINN) framework as a surrogate model for pressurized cavity expansion problems in homogeneous and bilayered isotropic and linear-elastic host media, with arbitrary boundary conditions and constitutive parameters. The proposed model combines known governing equations with observational data, which ensures physical consistency while providing mesh-free, high-fidelity solutions for both homogeneous and bi-layered media. FEM simulations were taken as ground truth and prediction results demonstrate robust predictive accuracy across heterogeneous domains. The proposed PINN significantly accelerates the prediction process, achieving up to 160 times faster computation compared to FEM while maintaining competitive accuracy, though discrepancies in shear stress and boundary errors persist due to the non-convexity of the loss function. These challenges can be mitigated by refining the weights of the loss function terms, improving the network architecture, and using advanced optimization techniques.

## 1 INTRODUCTION

Understanding the perturbations in physical fields induced by cavity expansion is essential for a broad spectrum of subsurface geotechnical engineering applications, particularly those where direct in-situ measurements are either impractical or economically prohibitive, ranging from subsurface exploration to deep tunnel boring (Bruland, 2000), enhanced geothermal systems (Henley and Ellis, 1983), carbon geosequestration and borehole drilling (Ali et al., 2022). Additionally, mechanical models of cavity expansion are indispensable to gain insights into bio-inspired burrowing phenomena, such as the growth mechanics of plant roots (Anselmucci et al., 2021), and to emulate the optimized excavation strategies demonstrated by burrowing organisms (Patino-Ramirez et al., 2020; Belachew et al., 2024).

Heterogeneous and anisotropic rock masses are most commonly encountered due to the pervasive distribution of discontinuities within the subsurface. And among these,

transversely isotropic rock masses are the most typical, especially in sedimentary and metamorphic formations (Song et al., 2023). Due to the variability and inherent directional dependency in constitutive parameters, and due to the prevalence of discontinuities, complex stress and displacement fields arise in heterogeneous rock masses. Most analytical solutions of cavity expansion problems are formulated as one-dimensional boundary value problems, and limited to idealized subsurface conditions, whereby the host medium is often assumed homogeneous and/or isotropic, which is not representative of realistic geological conditions.

Numerical methods, such as the Finite Element Method (FEM), make it possible to solve the stress and displacement fields around cavities of complex shapes embedded in non-elastic, anisotropic, and heterogeneous media. However, non-linear FEM simulations with large numbers of degrees of freedom are computationally demanding. This challenge becomes particularly pronounced in highly

non-linear inverse problems in which the FEM is used to back-calculate constitutive properties and reconstruct the response of the host medium from sparse stress or displacement values collected *in situ*, for which, running large numbers of simulations rapidly becomes impractical. In this context, Machine Learning (ML) emerges as a promising technique for surrogate modeling of cavity expansion. Once the ML model is trained, it can be used to bypass numerical methods and accelerate computations. However, conventional inductive or generative ML mapping inputs to outputs or learning data distribution from observed instances can hardly generalize to unseen instances. This limitation is alleviated in Physics-Informed Neural Networks (PINNs), in which governing equations are directly embedded into the objective function as physical constraints and penalties. Rather than solving large-scale linear systems with numerous degrees of freedom, PINNs approximate solutions that satisfy PDE constraints, and function as meshless partial differential operators, which ensures high-precision computation in local regions while enforcing compliance with physical laws. In this study, we propose a hybrid PINN framework to create a surrogate model for pressurized cavity expansion problems in homogeneous and bilayered media under arbitrary boundary conditions and constitutive properties. This novel meshless numerical framework enables the accurate and efficient reconstruction of physical responses in localized regions around cavities. Section 2 presents a brief review of the related literature. Section 3 explains the methodology adopted for data generation and PINN design. Section 4 summarizes the results obtained for the estimation of field variables around a circular pressurized cavity embedded in a homogeneous medium. Section 5 discusses the results obtained when training a PINN framework for estimating the stress and displacement fields around a cavity embedded in a bilayered geological host medium. Conclusions are presented in 6.

## 2 SCIENTIFIC BACKGROUND

The superior pattern recognition, generalization capabilities and computational efficiency of machine learning, compared to traditional numerical methods, have been extensively evaluated by researchers and leveraged for surrogate modeling of physical fields in diverse applications such as structural stress analysis, ground stress prediction, composite design optimization and biomechanics, to name only a few. For example, Liang et al. (2018) evaluated the feasibility of employing ML techniques as surrogates for finite element analysis to model the mechanical behavior of the human thoracic aorta wall. The proposed encoder-

decoder multi-subtask framework achieved high accuracy and fast Von Mises stress field prediction directly from the aorta wall geometry, but the model assumed that the tissue was homogeneous and isotropic with fixed constitutive parameters and residual stresses were not considered. Bolandi et al. (2022) trained a Convolutional Neural Network (CNN) to predict high-resolution stress contours in steel plates under various loading conditions and geometries. This ML algorithm was inherently limited to predicting stress contours in homogeneous materials, despite the fact that discontinuities, defects, and inclusions are commonly present in materials and structures. To bridge this gap, Rezasefat and Hogan (2023) explored the use of ML for stress field analysis around irregular inclusions. They trained a U-Net model using datasets generated from finite element simulations and evaluated the model performance through both quantitative and qualitative comparisons. However, the generalizability remained limited to specific scenarios represented within the training set. Thus far, most research in this area has focused on the reconstruction of physical fields in linear elastic materials. Khorrami et al. (2023) utilized a U-Net model as a surrogate for simulating heterogeneous polycrystalline ensembles of viscoplastic grains. The generalizability of the model was evaluated across various numbers of grains, grain morphologies, and extension histories not included in the training dataset. The proposed deep learning model achieved computational speeds 500 times faster than the counter-part numerical method used for forward-calculations. Furthermore, a more generalized model was developed to optimize high-performance composite materials, offering an efficient alternative to brute-force FEM sensitivity analyses. Yang et al. (2021) proposed a game-theory-based conditional Generative Adversarial Neural network (cGAN) to predict physical fields directly from the geometry of heterogeneous material microstructures. The model also enabled the derivation of secondary mechanical properties of the composites, such as mechanical recoverability. Subsequent experiments demonstrated that this robust framework could accurately capture the mechanical behavior and local patterns of materials, regardless of their geometries, shapes, loading conditions, and regardless of input image resolution. Jiang et al. (2021) proposed a cGAN framework for predicting the Von Mises stress field in two-dimensional solid structures. The input to the model consisted of three-channel images representing arbitrary geometries, boundary conditions, and loading conditions. The framework was trained and evaluated on datasets from fine-mesh structures and coarse-mesh cantilever beams. Results demonstrated that the cGAN framework significantly outperformed baseline models across multiple evaluation metrics. Gener-

alization experiments were also conducted, showcasing the framework's ability to handle unseen configurations effectively. While cGANs have revolutionized the field owing to their remarkable generalizability for stress field prediction, challenges such as training instability, non-convergence and generator collapse remain major obstacles to the deployment of cGAN networks as reliable FEM surrogates. To address these limitations, Denoising Diffusion Probabilistic Models (DDPMs) were introduced for mechanical analysis of physical fields. DDPMs demonstrated superior performance over cGAN networks, because they learn a reverse process for denoising, enabling more accurate and robust data generation. Jadhav et al. (2023) developed a conditional denoising diffusion-based model, StressD, to predict 2D stress fields in structures using channels of geometry, boundary conditions, and loading conditions. StressD is made of two modules. In the first one, the normalized stress field is predicted using a DDPM. In the second one, the maximum and minimum values of the stress field are predicted with a vision transformer. The normalized stress field predicted by the DDPM module is then rescaled accordingly. The framework was evaluated on fine-mesh, sparse-mesh, and hyperelastic datasets, demonstrating significant reductions in computational cost compared to the FEM while maintaining high accuracy. Buehler (2023) introduced a novel progressive attention-based transformer diffusion model for predicting 2D Von Mises stress fields near cracks. The model consists of a series of three U-Net architectures, which are trained individually and sequentially on a small dataset. Although trained on multi-crack datasets, the proposed model demonstrated exceptional generalizability when applied to unseen single-crack and distinct geometry scenarios. The results suggest that the model not only effectively captures the underlying principles of fracture mechanics, but also holds promise for predicting fracture propagation. The approaches reviewed above predominantly employ purely data-driven frameworks, which heavily depend on the quality of the training set, and exhibit poor transferability. Due to their purely mathematical structure, devoid of physical meaning, most machine learning algorithms can achieve impressive performance when provided with high-quality labels and large datasets. However, they generally struggle to extrapolate physical fields beyond the instance space.

PINNs were first introduced (Raissi et al., 2019) to explicitly incorporate partial differential equations (PDEs) into the objective function of data-driven methods as physical constraints. PINNs have since been widely adopted for solving PDEs as forward solvers, hence leveraging the universal approximation capability of deep neural net-

works as nonlinear function approximators, without the need for extensive training data (Hornik et al., 1989). A PINN can be viewed as a meshless approach that utilizes discretized residual points instead of solving equilibrium, kinematics, and constitutive equations into weak form discretized over finite elements. A PINN-based multi-network surrogate modeling framework for elastoplastic solid mechanics was introduced (Haghighat et al., 2021), in which governing equations are integrated directly into the loss function. In this approach, residuals are computed across the domain, boundary, and initial conditions, leveraging the spatial coordinates as network inputs. The framework demonstrated versatility in solving both linear elastic and perfectly elastoplastic solid mechanics problems while also solving inverse problems to estimate constitutive parameters. To extend the capabilities of PINNs, a framework tailored to finite-strain elastoplasticity was proposed (Niu et al., 2023) which incorporates rate-independent isotropic hardening into its formulation. The constitutive behavior was learned by a sub-network using discrete measurements from the stress-strain curve. The predicted outputs, including displacement, Cauchy stress, and accumulated plastic strain, were subsequently compared with results obtained from finite element analysis. Furthermore, a PINN was proposed for solving the governing equations of a non-associative Drucker-Prager elastoplastic solid with linear isotropic hardening rule (Roy et al., 2024) and the model performance was demonstrated by solving benchmark problems. Moreover, a spatial-time decomposition based on extended PINNs (XPINNs) was proposed (Jagtap and Karniadakis, 2020) for solving nonlinear partial differential equations on arbitrary complex geometry domains. The majority of existing PINN frameworks were designed to function as PDE operators that require retraining for each specific problem or set of conditions, addressing problems with fixed boundary conditions and predefined constitutive parameters.

The work presented in the following explores the application of PINNs as surrogate models for pressurized cavity expansion problems in homogeneous or heterogeneous media under arbitrary boundary conditions and with arbitrary constitutive properties, which, to the best of the authors' knowledge, no prior research has been investigated before.

### 3 METHODOLOGY

#### 3.1. Governing equations of linear elasticity

For a homogeneous, isotropic, and linear elastic solid under the assumption of small strain, the combination of constitutive relations, kinematic relations, and equilibrium equations gives the following Navier-Cauchy governing

equations in  $\Omega \in \mathbb{R}^2$ :

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = \nabla((\lambda + \mu) \nabla \cdot \mathbf{u}) + \nabla \cdot (\mu \nabla \mathbf{u}) + \rho \mathbf{f}_b, \quad (1)$$

where  $\mathbf{u}$  is the displacement field,  $\mathbf{f}_b$  denotes the body force,  $\rho$  is the mass density, and  $\lambda$  and  $\mu$  are the Lamé's coefficients.

### 3.2. PINNs: Physics-informed neural networks

A Physics-informed neural network  $\mathcal{NN}(\mathbf{X}; \tilde{\Theta})$  has the architecture of a fully connected neural network, comprising an input layer, an output layer, and multiple hidden layers. Here,  $\mathbf{X}$  represents the input vector, and  $\tilde{\Theta}$  denotes the optimal model weights.

While typically, PINNs use Cartesian coordinates and time as inputs, here, the model input,  $\mathbf{X} := (x, y; E, \nu; \sigma_v, k, P_c) \in \mathbb{R}^7$ , includes Cartesian coordinates, boundary conditions, and constitutive parameters.  $\sigma_v$  denotes the vertical stress applied in the far field,  $k$  denotes the horizontal-to-vertical far-field stress ratio (coefficient of Earth pressure), and  $P_c$  is the pressure (normal stress) applied at the cavity wall. The PINN approximates the displacement and stress fields, as follows:

$$(\hat{u}, \hat{v}, \hat{\sigma}_{xx}, \hat{\sigma}_{yy}, \hat{\sigma}_{xy}) = \mathcal{NN}(\mathbf{X}; \tilde{\Theta}), \quad (2)$$

The forward propagation through the hidden layers in a fully connected network can be expressed as:

$$\hat{y}^i = \Sigma^i(\hat{x}^{i-1}) := \sigma^i(W^i \hat{x}^{i-1} + b^i), \quad i = 1, 2, \dots, L, \quad (3)$$

where  $i$  denotes the index of the current layer,  $\hat{x}^{i-1}$  represents the output of layer  $L_{i-1}$  and serves as the input to layer  $L_i$ ,  $\sigma^i$  is the activation function applied at layer  $L_i$ ,  $W^i$  is the weight matrix for layer  $L_i$ , and  $b^i$  is the bias vector associated with layer  $L_i$ . The forward propagation through the entire network yields:

$$\mathbf{y} = \sigma^L(W^L \sigma^{L-1}(W^{L-1} \dots \sigma^1(W^1 \mathbf{x} + b^1) \dots + b^{L-1}) + b^L), \quad (4)$$

where  $\mathbf{y}$  represents the final output vector,  $\mathbf{x}$  denotes the input vector, and  $L$  is the total number of layers in the network.

The weights used in the PINN are optimized by iteratively reducing the value of the loss function using gradient descent. The gradient descent update rule, which adjusts the weights in the direction of the greatest descent of the loss, can be explained as follows:

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}), \quad (5)$$

where  $\mathbf{w}^{(t)}$  and  $\mathbf{w}^{(t+1)}$  denote the weights at the  $t$ -th and  $(t+1)$ -th iterations, respectively.  $\eta$  is the learning rate,

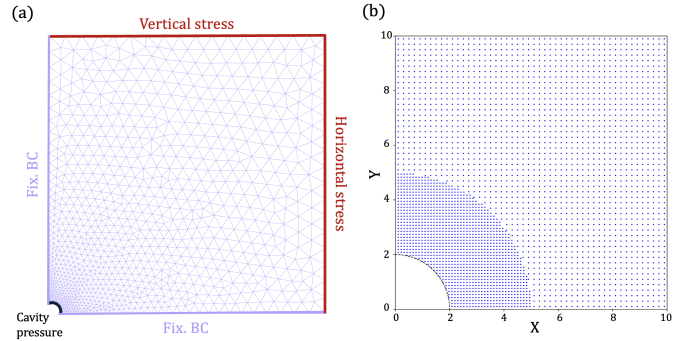


Fig. 1: (a) Schematic of the model setup with fixed displacements along the left and bottom sides, vertical compression stress ( $\sigma_v$ ) applied at the top and horizontal compression stress ( $k\sigma_v$ ) applied at the right. (b) Distribution of residual points within the region surrounding the cavity.

and  $\nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w})$  represents the gradient of the loss function  $\mathcal{L}(\mathbf{w})$  with respect to the weights  $\mathbf{w}$ . The ultimate goal of this process is to minimize the loss function by finding the optimal weights and converge to the optimal solution  $\tilde{\Theta}$ , as follows:

$$\tilde{\Theta} = \underset{\Theta}{\operatorname{argmin}} \mathcal{L}(\Theta). \quad (6)$$

### 3.3. Data generation

#### 3.3.1 Dataset for homogeneous media

Finite element simulations were used as the ground truth and served as a label to guide the PINN operator in approximating optimal solutions within the infinite solution space. The simulations were generated using the open-source finite element framework MOOSE (Giudicelli et al., 2024). The boundary conditions and geometry of a pressurized cavity in a homogeneous host medium are depicted in Fig. 1a. Due to the symmetry of the problem, only one-quarter of the domain was modeled, with a side length of 50 m and a cavity radius of 2 m. Arbitrary boundary conditions and constitutive parameters were selected randomly from Table 1. Residual points were sampled on a regular grid within the region near the cavity ( $10m \times 10m$ ), as illustrated in Fig. 1b. The sampling density was finer near the cavity and decreased progressively further from the cavity to capture the localized effects more effectively. In total, 3,000 simulations were conducted for different combinations of boundary conditions and material properties. The dataset was divided as follows: 70% for training, 20% for validation, and 10% for testing.

#### 3.3.2 Dataset for bilayered media

Cavity expansion was also simulated in a host medium made of two different homogeneous layers of isotropic linear elastic media. The boundary conditions were the

Table 1: Parameter ranges for simulations.

| Parameter (Symbol)                       | Range            |
|------------------------------------------|------------------|
| Vertical Stress ( $\sigma_v$ )           | [5 MPa, 50 MPa]  |
| Horizontal/Vertical Stress Ratio ( $k$ ) | [0.8, 1.5]       |
| Young's Modulus ( $E$ )                  | [15 GPa, 50 GPa] |
| Poisson's Ratio ( $\nu$ )                | [0.1, 0.25]      |
| Cavity Pressure ( $p_c$ )                | [0, 10 kPa]      |

same, but the domain was divided into two subdomains, each with different constitutive parameters, as illustrated in Fig. 2. The random material properties for both subdomains were selected from Table 1. The dataset was also split into three subsets with 70% for training, 20% for validation, and 10% for testing.

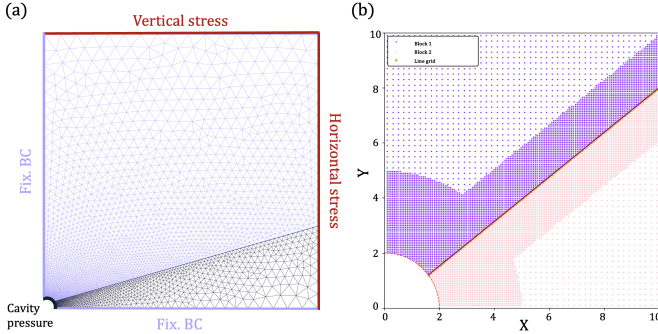


Fig. 2: (a) Schematic of the model setup for a bilayered medium with boundary conditions identical to those used for the homogeneous medium. (b) Distribution of residual points in the region surrounding the cavity.

## 4 PRESSURIZED CAVITY EXPANSION IN A HOMOGENEOUS MEDIUM

### 4.1. PINN structure

The architecture of the PINN  $\mathcal{NN}(\mathbf{X}; \tilde{\Theta})$  used to estimate the stress and displacements fields around a pressurized cavity embedded in a homogeneous medium is illustrated in Fig. 3. The network consists of 3 hidden layers, each containing 200 neurons. The Adam optimizer was used for training, with an initial learning rate set to  $5 \times 10^{-4}$ . A learning rate decay was applied every 500 epochs with a decay factor ( $\gamma$ ) of 0.95. The outputs of each hidden layer and the output layer were non-linearly transformed using the Tanh activation function.

### 4.2. Loss function

The loss function for the homogeneous case was defined as the weighted mean square error (MSE) derived from the kinematic relation, constitutive relation  $\mathcal{L}_c$ , equilibrium equation  $\mathcal{L}_e$  and observed instances  $\mathcal{L}_{data}$ . The

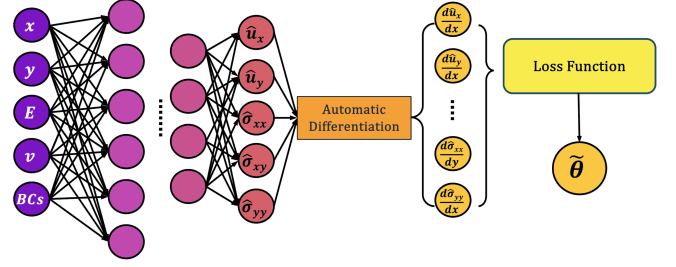


Fig. 3: Architecture of the PINN used for estimating displacement and stress fields around a pressurized circular cavity embedded in a homogeneous medium.

proposed numerical framework integrates mapping relations between perturbations and mechanical responses with physical constraints. The kinematic relationship is implicitly defined by the predicted displacement field, as given in Eq. (7),

$$\boldsymbol{\varepsilon} = \nabla^s \hat{\mathbf{u}} := \frac{1}{2} [\nabla \hat{\mathbf{u}} + \nabla \hat{\mathbf{u}}^T], \quad (7)$$

where  $\nabla^s$  denotes the symmetric gradient operator, and  $\hat{\mathbf{u}}$  represents the predicted displacement vector. The kinematic relation above is formulated under the small strain assumption.

The loss function is expressed in Eq. (8), where  $N_A$  denotes the number of collocation points  $\mathbf{X}_A \in \Omega$ . Here,  $\hat{y}$  represents the predicted values, and  $y^*$  denotes the ground truth.

$$\begin{aligned} \mathcal{L} &= \alpha_c \mathcal{L}_c + \alpha_e \mathcal{L}_e + \alpha_{data} \mathcal{L}_{data} \\ &= \frac{\alpha_c}{N_A} \sum_{A=1}^{N_A} |(\lambda + 2\mu)\varepsilon_{xx} + \lambda\varepsilon_{yy} - \hat{\sigma}_{xx}|^2 \\ &\quad + \frac{\alpha_c}{N_A} \sum_{A=1}^{N_A} |(\lambda + 2\mu)\varepsilon_{yy} + \lambda\varepsilon_{xx} - \hat{\sigma}_{yy}|^2 + \frac{\alpha_c}{N_A} \sum_{A=1}^{N_A} |2\mu\varepsilon_{xy} - \hat{\sigma}_{xy}|^2 \\ &\quad + \frac{\alpha_e}{N_A} \sum_{A=1}^{N_A} (|\hat{\sigma}_{xx,y} + \hat{\sigma}_{xy,x}|^2) + \frac{\alpha_e}{N_A} \sum_{A=1}^{N_A} (|\hat{\sigma}_{xy,y} + \hat{\sigma}_{yy,x}|^2) \\ &\quad + \frac{\alpha_{data}}{N_A} \sum_{A=1}^{N_A} |\hat{u}_x - u_x^*|^2 + \frac{\alpha_{data}}{N_A} \sum_{A=1}^{N_A} |\hat{u}_y - u_y^*|^2 \\ &\quad + \frac{\alpha_{data}}{N_A} \sum_{A=1}^{N_A} |\hat{\sigma}_{xx} - \sigma_{xx}^*|^2 + \frac{\alpha_{data}}{N_A} \sum_{A=1}^{N_A} |\hat{\sigma}_{yy} - \sigma_{yy}^*|^2 \\ &\quad + \frac{\alpha_{data}}{N_A} \sum_{A=1}^{N_A} |\hat{\sigma}_{xy} - \sigma_{xy}^*|^2. \end{aligned} \quad (8)$$

The validation loss was defined as the mean absolute error (MAE) between the ground truth and the predicted values, formulated as:

$$\mathcal{L}_{val} = \frac{\alpha_{data}}{N_A} \sum_{A=1}^{N_A} |\hat{y} - y^*|. \quad (9)$$

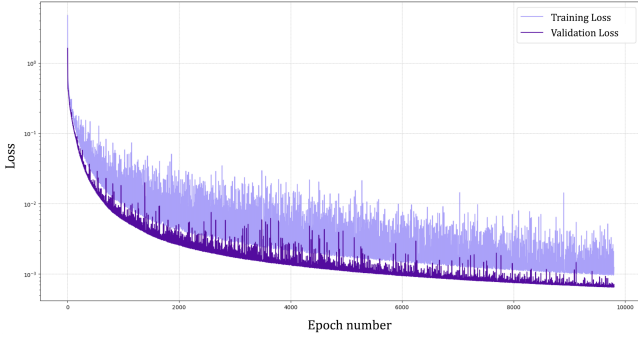


Fig. 4: Evolution of the training and validation loss functions with the number of epochs.

Following 9,800 epochs of training, all loss terms began to converge, as illustrated in Fig. 4.

### 4.3. Results and discussion

The performance of the model was evaluated on the test dataset. The comparison between the predicted physical fields and the ground truth is illustrated in Fig. 5 for a typical instantiation of the model. The displacement and stress fields are accurately forecast under arbitrary boundary conditions and with arbitrary constitutive parameters. Unlike traditional mesh-based or resolution-limited machine learning approaches where accuracy heavily depends on a predefined spatial discretization, this framework achieves local, high-precision, grid-free predictions, thereby capturing more complex mechanical responses near the cavity and adhering to the underlying physical laws, which also suggests that the framework could be potentially extended to more intricate cavity geometries. A small discrepancy was observed at the boundary  $\mathbf{X} \in \Gamma$  (see Fig. 5f and Fig. 5o), which may indicate the need to incorporate a Dirichlet boundary condition loss term into the objective function to better approximate the optimal solution in the hypothesis space. Stress concentration was clearly identified around the cavity, while critical errors predominantly occurred at residual points with lower stress concentrations, and fewer errors were found in regions exhibiting higher stress concentrations (Fig. 5i and Fig. 5l). Since the model inputs are normalized and the outputs are nonlinearly transformed by the Tanh activation function, minor numerical precision issues may arise, and the loss curve suggests that additional training epochs are required to further improve the model. Figure. 6 illustrates the error distribution across the entire test dataset. The proposed PINN surrogate model allows accurate physical field predictions and strong generalizability, maintaining robustness across varying geological conditions.

## 5 PRESSURIZED CAVITY EXPANSION IN A BI-LAYERED MEDIUM

### 5.1. PINN structure

A PINN framework is proposed for surrogate modeling of cavity expansion in bilayered media, using a domain decomposition method. The global solution domain  $\Omega := \cup_i^{N_d} \Omega_i$  is partitioned into non-overlapping subdomains  $\Omega_i$ , each of which is approximated by a dedicated sub-network operator. These sub-operators are then coupled via a continuity condition, which can be formulated as:

$$\mathbf{Y} = \mathcal{N}_{\Omega_i}(\mathbf{X}_{\Omega_i}; \tilde{\Theta}_{\Omega_i}), \quad (10)$$

where  $\mathbf{Y}$  denotes the predicted physical fields,  $\mathbf{X}_{\Omega_i}$  represents the input field variables, boundary conditions and constitutive parameters for subdomain  $\Omega_i$ , and  $\tilde{\Theta}_{\Omega_i}$  is the optimal parameters of the corresponding neural network. The structure of the proposed heterogeneous PINN is illustrated in Fig. 7. As shown in the figure, the input vector differs from the homogeneous case, as all residual points within the domain are encoded with their respective coordinates, boundary conditions  $B_{\Omega_i}$ , and constitutive properties from both subdomains  $\sum P_{\Omega_i}$ . This encoding enables the model to adaptively perceive and dynamically capture the mechanical responses of adjacent computational domains. Each sub-network consists of three hidden layers, with each layer containing 150 neurons. Distinct activation functions were employed for different sub-networks (Jagtap and Karniadakis, 2020), with Tanh used for  $\Omega_1$  and GeLU for  $\Omega_2$ . The Adam optimizer was employed for training, with an initial learning rate of  $1 \times 10^{-3}$ . Additionally, a Cosine Annealing learning rate schedule was implemented to prevent overfitting, applying a decay factor ( $\gamma$ ) of 0.95 every 1,000 epochs, with a learning rate varying between a maximum of  $1 \times 10^{-3}$  and a minimum of  $1 \times 10^{-4}$ .

### 5.2. Loss function

To ensure the continuity of displacements at the boundary between the two subdomains, an additional term  $\mathcal{L}_s$  is introduced in the loss function expressed in Eq. 8, as follows:

$$\begin{aligned} \mathcal{L} &= \sum_{i=1}^j (\alpha_{c_i} \mathcal{L}_{c_i} + \alpha_{e_i} \mathcal{L}_{e_i} + \alpha_{data_i} \mathcal{L}_{data_i}) + \sum_{i=1}^{j-1} \alpha_{s_i} \mathcal{L}_{s_i} \\ &= \sum_{i=1}^j (\alpha_{c_i} \mathcal{L}_{c_i} + \alpha_{e_i} \mathcal{L}_{e_i} + \alpha_{data_i} \mathcal{L}_{data_i}) + \sum_{i=1}^{j-1} \frac{\alpha_{s_i}}{N_s} |\hat{\sigma}_{xy_1} - \hat{\sigma}_{xy_2}|^2 \\ &\quad + \sum_{i=1}^{j-1} \frac{\alpha_{s_i}}{N_s} |\hat{\sigma}_{yy_1} - \hat{\sigma}_{yy_2}|^2 + \sum_{i=1}^{j-1} \frac{\alpha_{s_i}}{N_s} |\hat{\sigma}_{xx_1} - \hat{\sigma}_{xx_2}|^2 \\ &\quad + \sum_{i=1}^{j-1} \frac{\alpha_{s_i}}{N_s} |\hat{u}_{x_1} - \hat{u}_{x_2}|^2 + \sum_{i=1}^{j-1} \frac{\alpha_{s_i}}{N_s} |\hat{u}_{y_1} - \hat{u}_{y_2}|^2, \end{aligned} \quad (11)$$

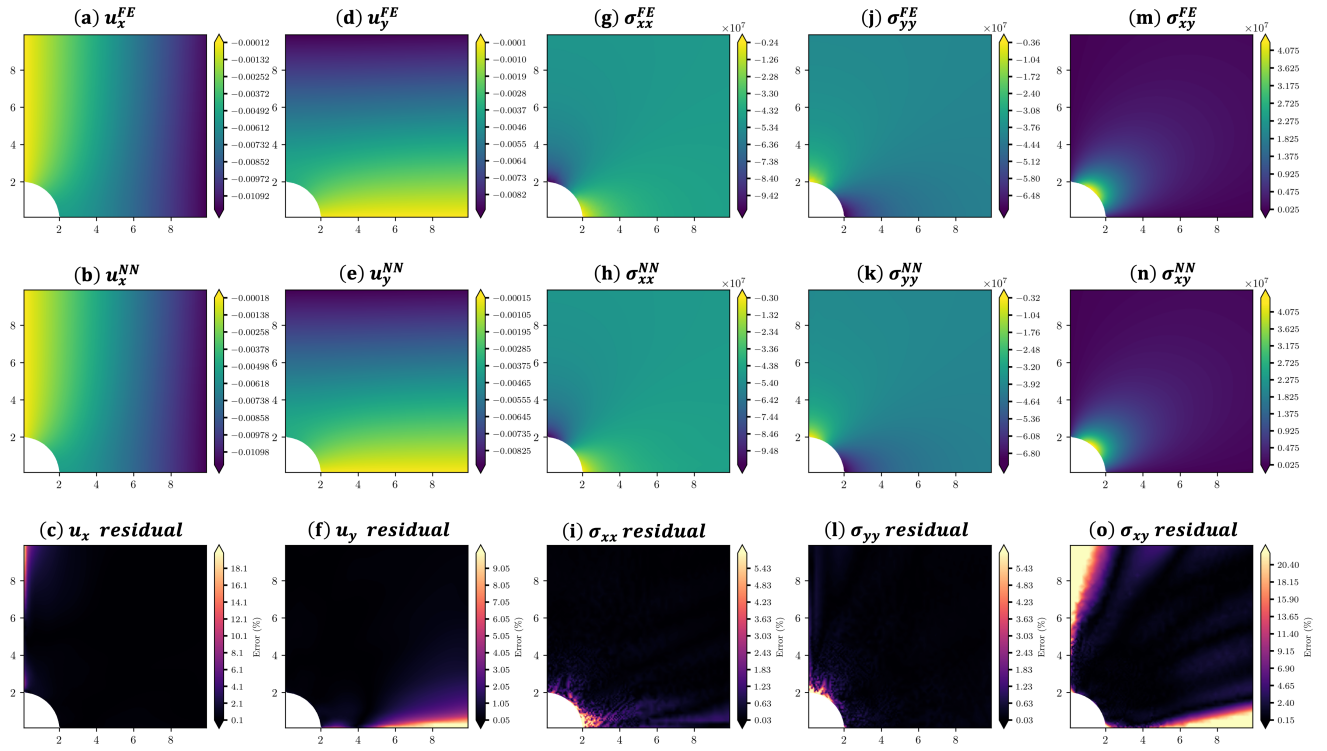


Fig. 5: Comparison between ground truth and predicted data for the problem of pressurized cavity in a homogeneous medium, for an example instantiation of the boundary conditions and constitutive parameters. The first row represents data generated by FEM, the second row shows the predicted fields, and the third row depicts the relative error between them.

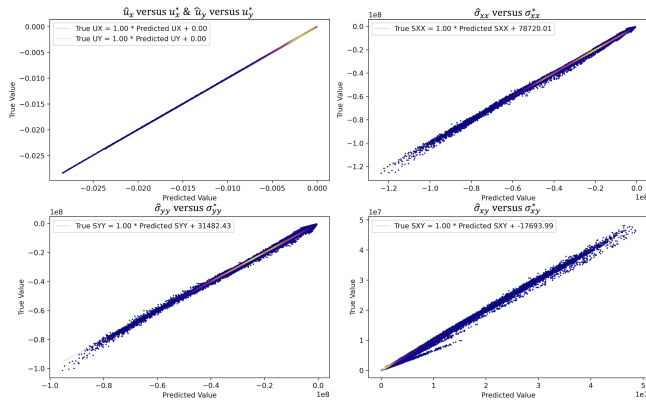


Fig. 6: Ground truth vs. predicted data displacement and stress components across the entire test data set used to study cavity pressurization in a homogeneous medium.

where  $N_s$  represents the number of residual points along the boundary between the two subdomains. This formulation enforces alignment with the continuity condition at the boundary between the two subdomains, enhancing the stability and accuracy of the PINN used to simulate cavity pressurization in a bilayered medium.

### 5.3. Results and discussion

The comparison between the ground truth and the predicted physical fields is shown in Fig. 8 for an example

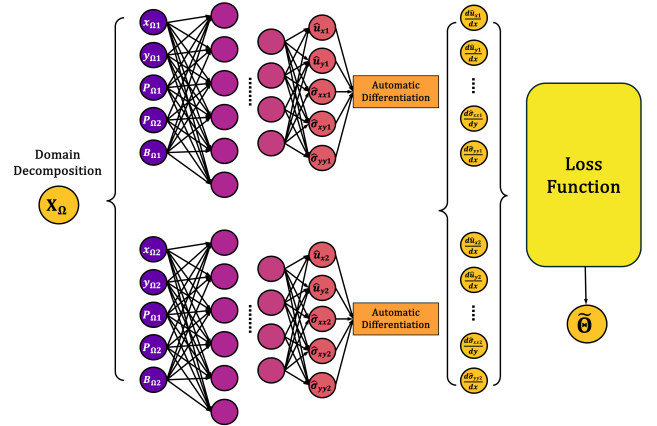


Fig. 7: Architecture of the PINN used for estimating displacement and stress fields around a pressurized circular cavity embedded in a bilayered medium.

model instantiation. The training and validation loss functions after 8,000 epochs are presented in Fig. 9, and the overall predictions on the test set are presented in Fig. 10. The model successfully captures both the displacement and stress fields, with the exception of shear stress. The largest displacement errors are observed near the domain boundaries, whereas the most significant stress field errors occur in low-stress regions near the cavity. Notably, the proposed PINN fails to learn the shear stress distribution

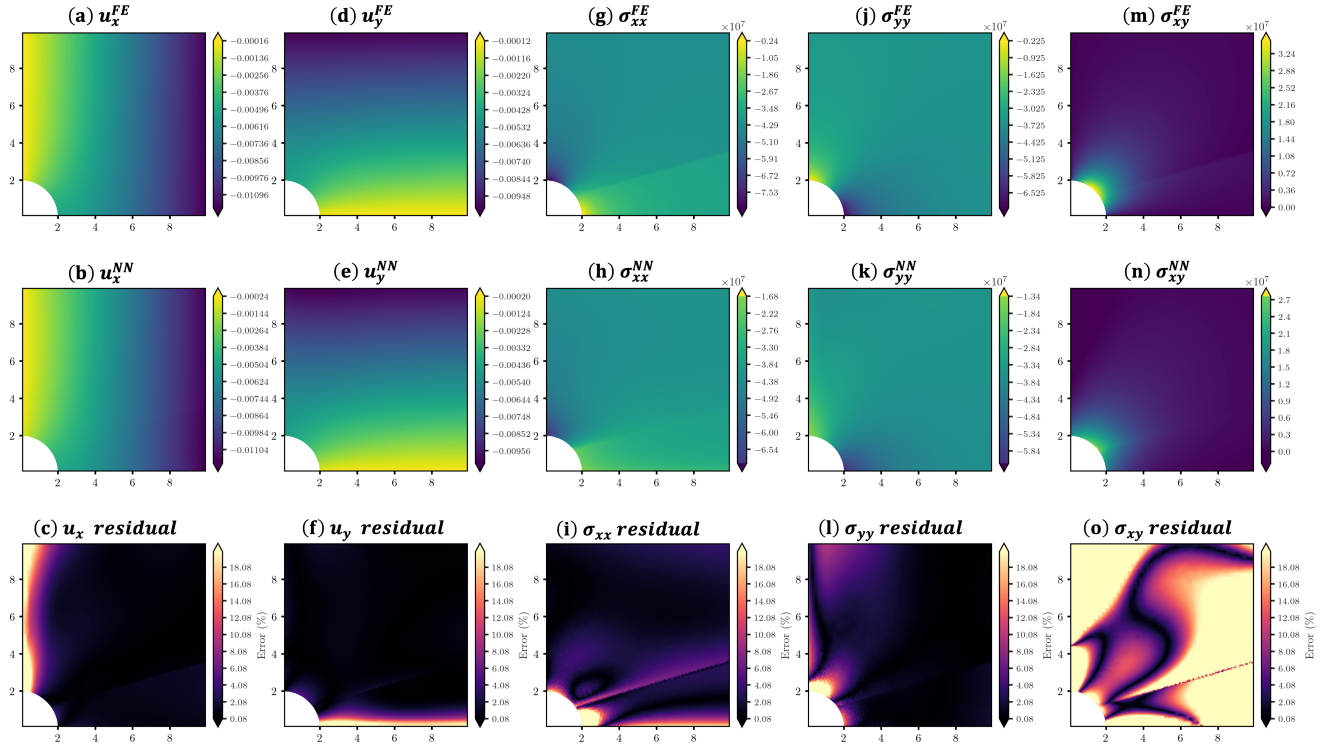


Fig. 8: Comparison between ground truth and predicted data for the problem of pressurized cavity in a bilayered medium, for an example instantiation of the boundary conditions and constitutive parameters. The first row represents data generated by FEM, the second row shows the predicted fields, and the third row depicts the relative error between them.

across the two domains, and the displacement estimations exhibited greater errors compared to the case of a cavity embedded in a homogeneous medium. This discrepancy may be attributed to the inclusion of an excessive number of loss terms, which might lead to convergence difficulties. The resulting loss function may become non-convex, increasing the likelihood of the model getting trapped in local minima. Furthermore, Fig. 9 suggests that the training process has not yet fully converged, indicating potential for further improvement.

To assess the computational efficiency of the proposed hybrid PINN framework, the computation time required to generate 100 predictions was compared against that of the finite element method, as summarized in Table 2. In the bilayered media case, FEM exhibited slightly lower computational time than in the homogeneous case, primarily due to smaller mesh complexity and degrees of freedom. The proposed hybrid PINN framework demonstrated significantly superior computational performance while maintaining high predictive accuracy. Specifically, it achieved 100 predictions in 2.76s for homogeneous media and 3.12s for heterogeneous media, representing a speedup of up to 160x compared to FEM. This remarkable acceleration highlights the framework's capability to efficiently deliver high-fidelity, mesh-independent solu-

tions under arbitrary boundary conditions and constitutive parameters. Such performance underscores the potential of the proposed PINN approach as an efficient surrogate model for pressurized cavity expansion problems, offering a promising tool for real-time analysis and large-scale predictive modeling in geotechnical applications, and beneficial to a broad range of downstream tasks.

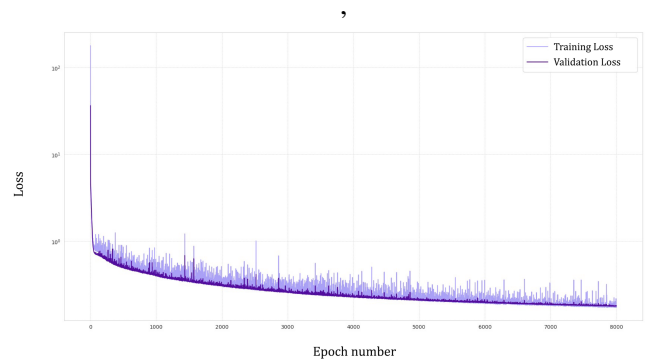


Fig. 9: Loss terms of the PINN proposed for bilayered media.

To improve accuracy, particularly in displacement field prediction and shear stress approximation, future work will explore deeper network architectures and advanced optimization strategies to enhance training stability and convergence.

Table 2: Comparison of total computation time for predicting 100 samples using FEM and PINN in homogeneous and bilayered media.

| Method | Homogeneous   | Heterogeneous |
|--------|---------------|---------------|
| FEM    | 439.25 s      | 291.41 s      |
| PINN   | <b>2.76 s</b> | <b>3.12 s</b> |

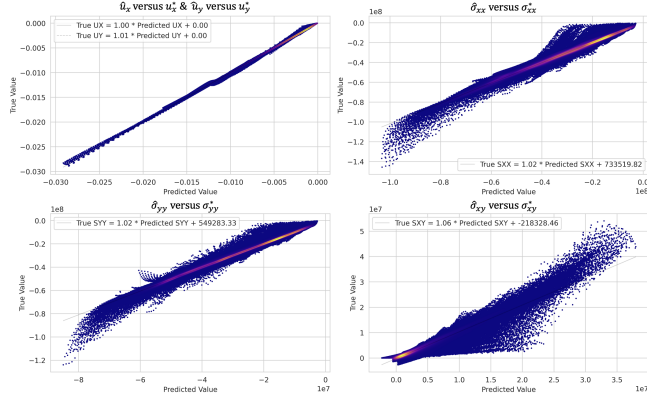


Fig. 10: Ground truth vs. predicted data displacement and stress components across the entire test data set used to study cavity pressurization in a bilayered medium.

## 6 CONCLUSIONS

In this research, we proposed a novel physics-informed neural network (PINN) framework as a surrogate model for approximating solutions to the circular pressurized cavity expansion problem in homogeneous and bilayered media, under arbitrary boundary conditions and with arbitrary isotropic linear elastic constitutive properties. This framework demonstrates robust and generalized field prediction capabilities across various rock formations and geostress conditions, even in topologically complex or discontinuous domains. In addition to its strong predictive accuracy, the proposed PINN significantly improves computational efficiency. Once trained, it generates accurate and generalizable predictions at up to 160 times the speed of conventional FEM, enabling continuous and real-time mechanical response inference in geotechnical applications.

For both homogeneous and heterogeneous cases, the largest errors typically occur near the domain boundary, and the proposed PINN framework exhibits significant shear stress errors away from the cavity region. Such issues may arise from the non-convex nature of the multi-objective loss function, potentially causing the model to converge to local minima. These challenges can be mitigated through appropriate model tuning.

Overall, the proposed method provides a meshless, physics-informed alternative to traditional numerical

methods, integrating observed data with fundamental physical principles. It offers detailed local solutions in heterogeneous media, exhibiting considerable promise for advancing solid mechanics and earth science applications such as tunnel boring, geothermal systems, and carbon geosequestration.

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